

Aarav Pant

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SUMMARY

Fourth-year Computer Science student pursuing a career as a **Software Engineer specializing in AI/ML and Cloud Security**, with experience spanning 16 months of software engineering at Evertz, cloud security tooling, and 8 months as a Teaching Assistant at McMaster. Targeting CompTIA Security+ (Dec 2026), AWS SAA-C03 (Jan 2027), and Google Professional ML Engineer (Feb 2027).

EDUCATION

McMaster University

Hamilton, ON, Canada

Bachelor of Engineering in Computer Science (Honours, Co-op), Minor in Mathematics, **4th Year**

Sep 2022 – Apr 2027

Dean's Honour List, **GPA: 3.7/4.00**

- **Certifications:** CompTIA Security+ (SY0-701) (*In Progress — Target: Dec 2026*), AWS SAA-C03 (*In Progress — Target: Jan 2027*), Programming with JavaScript by Meta, Introduction to Front-End Development by Meta, Python for Data Science, AI & Development by IBM, Introduction to Web Development (HTML, CSS, JavaScript) by IBM, Version Control by Meta.
- **Relevant Coursework:** Discrete Math, Introduction to Programming, Intro to Software Using Web Programming, Automata and Computability, Data Structures & Algorithms, Databases, Computer Architecture, Intro to Software Development, Concurrent Systems, Algorithms and Software Design, Algorithms and Complexity, Data Privacy, Principles of Programming Languages, Computer Networks and Security, Operating Systems, Syntax-Based Tools & Compilers, Applied Cryptography, Software Testing, Fundamentals of Machine Learning, Linear & Continuous Optimization, Capstone Project.

SKILLS

Programming Languages: Python, Java, C, C++, C#, Kotlin, JavaScript, TypeScript, HTML, CSS, Haskell, Elm

Databases: MySQL, MongoDB, PostgreSQL, SQLite

Frameworks, Libraries, and Tools: React, React Native, Next.js, Redux, Tailwind CSS, Bootstrap, Express, Django, Three.js, PyTorch, TensorFlow, Keras, Hugging Face Transformers, LangChain, NumPy, Scikit-learn, Firebase, FastAPI, Matplotlib, Pandas, Docker, Kubernetes, Jenkins, Postman, JUnit, Git, GitHub, Jira

Cloud and DevOps: AWS (S3, EC2, Lambda, IAM, RDS, DynamoDB, SNS, EventBridge, CloudWatch), GCP (Cloud Run), Terraform, GitHub Actions, Docker Compose, CI/CD Pipelines

EXPERIENCE

McMaster University

Hamilton, Ontario, Canada

Teaching Assistant, **COMPSCI 2XC3 & CS 1MD3**

Jan 2025 – Apr 2025, Sept 2026 – Present

- Supported students in **CS 1MD3** on programming fundamentals and in **COMPSCI 2XC3** on algorithm design & analysis and software design principles across lab sections of **30–40 students**.
- Graded **150+ exams** and assisted with technical documentation and presentations, providing clear feedback to help students improve their problem-solving and coding skills.
- Invigilated midterm and final examinations alongside the course instructor, helping ensure a smooth, organized, and fair testing environment.

Evertz Microsystems Ltd.

Burlington, ON, Canada

Software Engineer Intern

May 2025 – August 2026

- Developed and executed comprehensive test cases for the **Mediator-X** platform, ensuring robust media asset management and playlist functionalities across diverse client environments.
- Collaborated directly with clients to gather requirements and deliver customized solutions, including dynamic reports, SQL queries, and automation scripts, enhancing operational efficiency.
- Engineered automation tools using Python and JavaScript to streamline Mediator-X workflows, cutting one script's runtime from **~2 min to under 1 min**.

Google Developer Student Club (GDSC) | [GitHub](#)

McMaster University, Hamilton, ON

Open Source Team Member (Software Developer)

Oct 2024 – April 2025

- Collaborated with **9 developers** to design and implement 3 key features for the DBAC Companion App using React Native and Tailwind CSS, enhancing usability for over 50 students.
- Redesigned the application interface, leading to a **10% increase in user satisfaction** and streamlining navigation workflows for a seamless user experience.
- Utilized Google Cloud Platform (**GCP**) for database management and Firestore for real-time data synchronization, ensuring efficient and scalable data handling.

PROJECTS

AWS Cloud Security Posture Scanner | *Python, boto3, Terraform, AWS, Kubernetes* | [GitHub](#)

- Built an automated auditing tool implementing **5 CIS AWS** and **4 CIS Kubernetes Benchmark checks**, validated live against a real AWS account and Kubernetes cluster.
- Designed a serverless scan pipeline (**Lambda + EventBridge + DynamoDB + SNS**) via Terraform with a findings lifecycle and opt-in auto-remediation for low-risk findings.
- Built a CloudWatch dashboard and custom metrics pipeline for scan latency and findings trends, backed by a least-privilege IAM role scoped per capability rather than one broad policy.

Technical Knowledge RAG Assistant | *FastAPI, PyTorch, Vector DB (Chroma), GCP Cloud Run, Docker* | [GitHub](#) | [Live Demo](#)

- **Built a RAG pipeline** over a multi-domain technical corpus (security, cloud architecture, software engineering) with GPU-accelerated embeddings and a locally-hosted LLM generating citation-grounded answers.
- **Implemented a structural hallucination guard** that rejects generation when retrieval confidence falls below an empirically-derived threshold, rather than relying on prompt instructions alone.
- Built a 39-question evaluation harness across 5 documents — **74.3% retrieval precision**, 80% keyword recall, **0% hallucination rate** — fixing 4 real bugs along the way.

Algorithmic Trading Signal Scanner | *Python, Flask, React, Interactive Brokers API* | [GitHub](#)

- **Built two independent rule-based scoring engines** (5-min intraday scanner, swing evaluator) that score live market data against a multi-factor confluence model (VWAP, RSI, OBV, relative strength) and log every signal for accuracy tracking.
- **Engineered a full-stack Flask + React dashboard** with an **Interactive Brokers** data layer and an **MCP server** for AI agents.
- **Built a walk-forward backtester**; used results to disable a losing signal mode (**PF 0.06**).